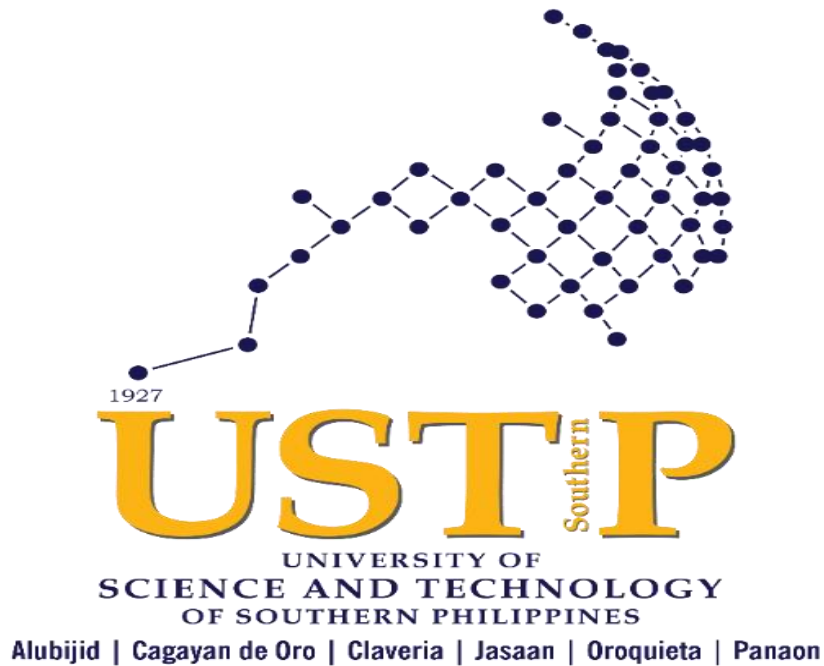


Case Study-IT314



Script4ce
LG John Labalan

Handog Bayan: A Community-Driven Donation and Relief Management System

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Abstract

Handog Bayan is a technology-enabled donation and relief management system designed to streamline the process of collecting, validating, and distributing donations to communities in need, particularly in the Philippines where disasters and poverty remain persistent challenges. Traditional donation drives are often manual, fragmented, and lacking transparency, which can result in delayed assistance, duplication of efforts, and reduced donor trust.

This case study presents the design and implementation of Handog Bayan as a mobile and web-based platform that connects donors, beneficiaries, and administrators in a unified environment. Using structured requirement elicitation techniques and an iterative development methodology, the system was modelled with clear use cases, process flows, and data structures. The study discusses the system architecture, key features (such as user registration, request posting, donation pledging, and status tracking), and the technologies used. Evaluation from initial users indicates that the system improves coordination, documentation, and transparency in donation and relief operations. Recommendations for future enhancements include integration with payment gateways, more advanced analytics, and partnerships with government and non-government organizations.

1. Introduction

1.1 Background of the Study:

The Philippines is highly vulnerable to natural disasters such as typhoons, floods, earthquakes, and volcanic eruptions, as well as chronic socio-economic issues like poverty and food insecurity. In response, individuals, schools, churches, and civic organizations regularly conduct donation drives and community outreach programs. However, these initiatives often rely on manual processes: posting calls for help on social media, collecting donations physically, tracking contributions using spreadsheets or notebooks, and coordinating distribution through informal networks.

These manual methods can lead to several issues:

- Lack of centralized information on who needs help, what kind of help is needed, and which organizations are already responding.
- Limited transparency for donors about how their contributions are used.
- Inefficient distribution, resulting in oversupply in some areas and shortages in others.
- Given the increasing accessibility of smartphones and the internet, there is an opportunity to create a digital platform that supports organized, transparent, and efficient donation management for communities. Handog Bayan is conceptualized as such a platform.

1.2 Statement of the Problem:

This study addresses the following problems:

- P1: Fragmented communication and coordination.
There is no centralized system to match donors with verified requests from beneficiaries or community organizations.

- P2: Inefficient tracking of donations and requests.
Current practices make it difficult to monitor donation status, inventory, and fulfillment of specific needs.
- P3: Limited transparency and documentation.
Donors often do not receive feedback or clear reports about the impact of their contributions, which can lower trust and willingness to donate again.
- P4: Lack of data for decision-making.
Organizers lack historical data and analytics to plan better for future relief operations and outreach projects.

1.3 Objectives of the Study:

General Objective

- To design and develop a digital system called Handog Bayan that improves the management, transparency, and coordination of donations and community assistance.

Specific Objectives

- To identify and analyze the needs of donors, beneficiaries, and administrators in the context of donation and relief operations.
- To design a system architecture and data model that support user registration, request creation, donation pledging, and status tracking.
- To implement a prototype of Handog Bayan using appropriate software technologies.
- To evaluate the system in terms of usability, functionality, and perceived effectiveness from selected users.
- To document the development process, system design, and lessons learned for future enhancement.

1.4 Significance of the Study:

- **Donors**
Gain a convenient and transparent way to see verified requests, pledge donations, and track the status of their contributions.
- **Beneficiaries / Community Requesters**
Obtain a formal channel to submit assistance requests, specify detailed needs, and receive timely support.
- **Administrators / Organizers**
Can verify requests, manage campaigns, match donations to needs, and generate reports for accountability and planning.
- **Educational Institutions and Researchers**
May use the case study as a reference for implementing community-centered digital solutions and for future academic research.
- **Government and NGOs**
May adapt or integrate similar systems to support larger-scale relief and social welfare programs.

1.5 Scope and Limitations of the Study:

Scope

- Focuses on a **community-level donation and relief management system**, primarily for non-monetary donations (e.g., food packs, clothes, hygiene kits, school supplies).
- Covers **three main user roles**: donors, beneficiaries (or requesters), and administrators.
- Includes basic features such as user registration/login, email or SMS verification, posting requests, viewing requests, pledging donations, updating statuses, and generating basic summary reports.

Limitations

- The initial version may not include integrated online payment or e-wallet functionality; monetary donations, if any, are handled externally.
- System evaluation is limited to a small sample of users (e.g., students, local community members, selected organizations).
- Infrastructure (e.g., internet connectivity) in rural or disaster-stricken areas may affect real-time system usage.
- The prototype may not yet be fully integrated with government disaster-response platforms or large NGO systems, though future integration is envisioned.

2. Related Literature

- **Digital Humanitarianism and Crowdsourcing in Disasters**
Meier (2015), in *Digital Humanitarians*, discusses how digital platforms and volunteers can collect, process, and visualize crisis data to support faster humanitarian responses. This emphasizes the role of technology and citizen participation in improving situational awareness, which supports the idea of Handog Bayan as a crowdsourced information hub for community needs and donations.
- **Promoting Digital Humanitarian Action in Protecting Human Rights**
A study published in the *Journal of International Humanitarian Action* highlights that digital humanitarian tools can strengthen the protection of vulnerable populations by enabling better coordination, transparency, and accountability in humanitarian operations. The authors note that the effectiveness of such systems depends on inclusive design and reliable data, both of which are key considerations in Handog Bayan.
- **Local Aid Workers in the Digital Humanitarian Project**
An article from *Critical Asian Studies* analyzes the experiences of local aid workers in digitally mediated disaster response operations, particularly following Typhoon Haiyan in the Philippines. It underscores the importance of local context, labor conditions, and meaningful participation of local actors. Handog Bayan addresses this by positioning local administrators and community leaders as primary users and decision-makers within the system.
- **Mobile Applications for Donation and Charity**
Various studies on mobile-based donation platforms have found that usability, trust, and transparency significantly influence users' willingness to donate. Systems that provide clear information about beneficiaries, real-time status updates, and feedback loops tend to foster stronger donor engagement. These findings guide the design of Handog Bayan's user interface and notification features.

3. Methodology

3.1 Requirement Elicitation Techniques and Overall Methodology

Requirement Elicitation Techniques

- **Key Informant Interviews**
Conducted with community organizers, social workers, and representatives from local organizations to understand current donation practices and pain points.
- **Surveys / Questionnaires**
Distributed to potential donors and beneficiaries to capture their expectations for a digital donation platform (e.g., preferred features, privacy concerns, notification preferences).
- **Observation**
Informal observation of actual donation drives or outreach activities to see how donations are collected, recorded, and distributed.
- **Document Analysis**
Review of existing records such as spreadsheets, forms, and social media posts used in previous donation drives to see how information is currently structured.
- **Use Case and Scenario Analysis**
Development of user stories and scenarios (e.g., “A donor wants to help a specific family,” “A barangay captain posts a request after a flood”) to refine functional requirements.

Development Methodology

- **Agile / Iterative Approach**
The system was developed in short iterations (sprints), with continuous feedback from representative users. Each iteration included:
 - Requirement refinement
 - Design (UML diagrams, wireframes)
 - Implementation (coding)
 - Testing and feedback

This approach allowed early detection of usability issues and progressive enhancement of features.

3.2 Software and Hardware Requirements

Software Requirements

- Operating System: Windows / macOS / Linux for development
- Development Framework: Cross-platform mobile framework (e.g., Flutter)
- Programming Language: Dart (for Flutter-based client), plus backend language if applicable (e.g., Node.js/Java/PHP)
- Database: Cloud-based NoSQL or SQL database (e.g., Firebase Firestore, MySQL, or PostgreSQL)
- Backend / Cloud Services: RESTful API or BaaS (e.g., Firebase Authentication, Cloud Functions)
- IDE/Tools: Android Studio, VS Code, Git for version control
- Diagramming Tools: Draw.io, Lucidchart, or similar

Hardware Requirements

- **Client Devices**
 - Android smartphone (minimum Android 8.0) or web browser for web version
 - At least 2 GB RAM
 - Stable internet connection
- **Server / Hosting (if self-hosted backend)**
 - Cloud server or hosting service
 - At least 2 vCPUs, 4 GB RAM, and sufficient storage for application and database
 - Regular backups and basic security configuration
- **Development Machine**
 - At least Intel i5 or equivalent
 - 8 GB RAM or higher
 - 256 GB storage

4. Design and Implementation

4.1 Architectural Design:

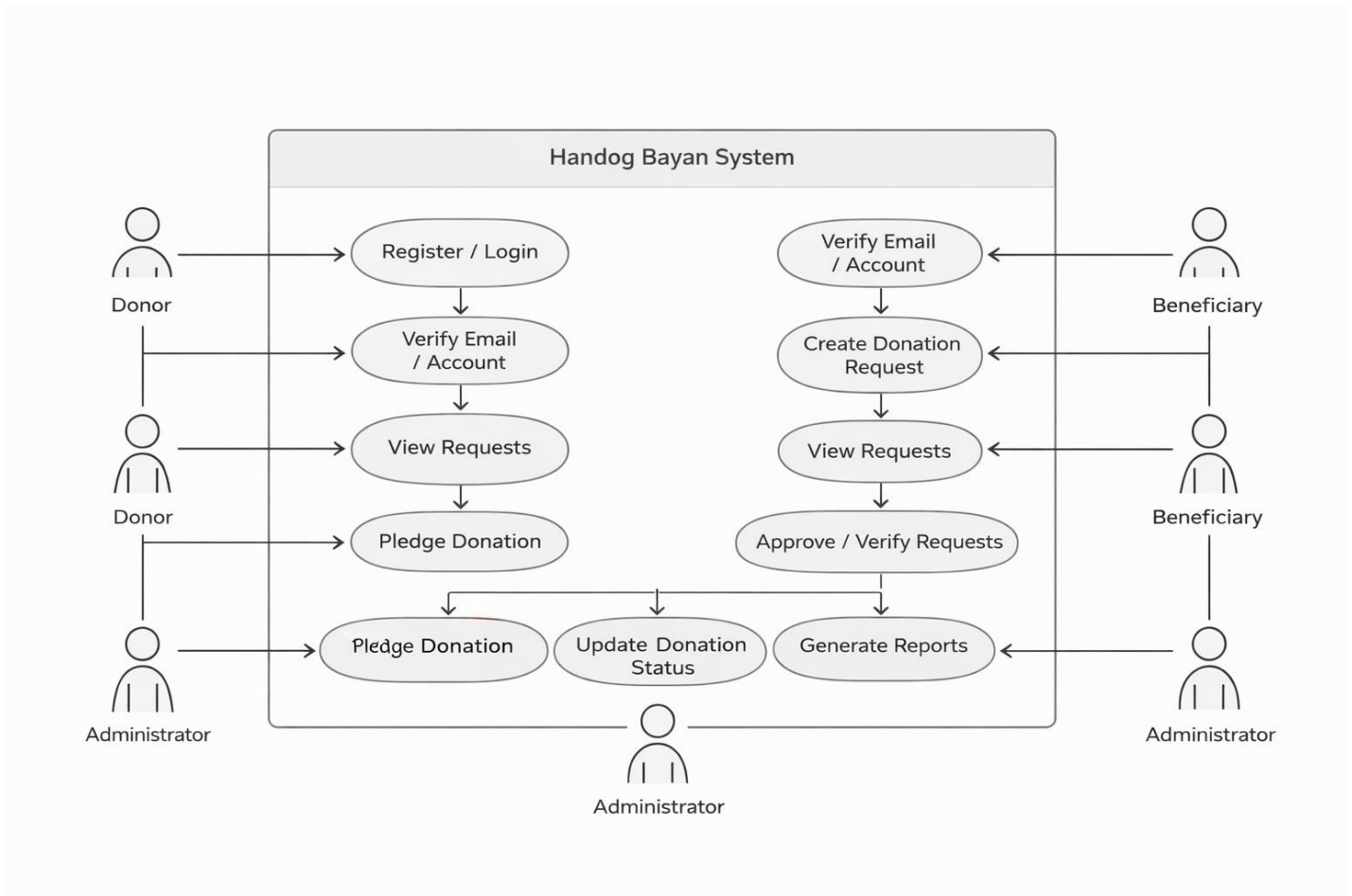
The Handog Bayan system follows a **three-tier architecture**:

- **Presentation Layer (Client)**
Mobile application (and/or web interface) that provides user interfaces for donors, beneficiaries, and administrators. Handles input validation and basic navigation.
- **Application / Service Layer (Backend)**
Exposes APIs for authentication, request management, donation matching, and reporting. Applies business rules such as validation of requests, status transitions, and role-based access control.
- **Data Layer (Database)**
Stores user accounts, beneficiary profiles, donation requests, donation pledges, inventory, and transaction logs. Ensures data integrity and supports reporting.

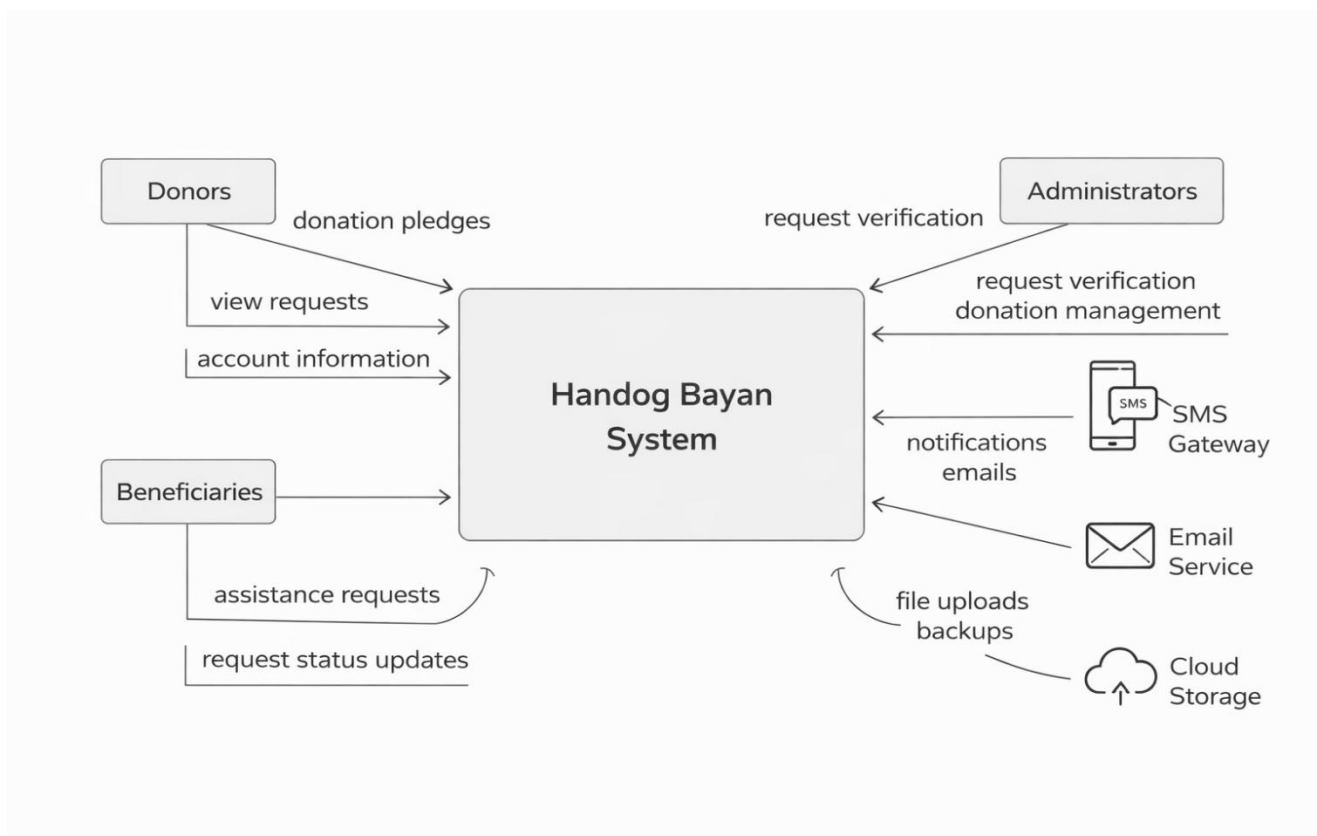
Data flows from the user's device (client) to the backend server, which processes requests and updates the database; responses are then sent back to the client for display.

4.2 Diagrams:

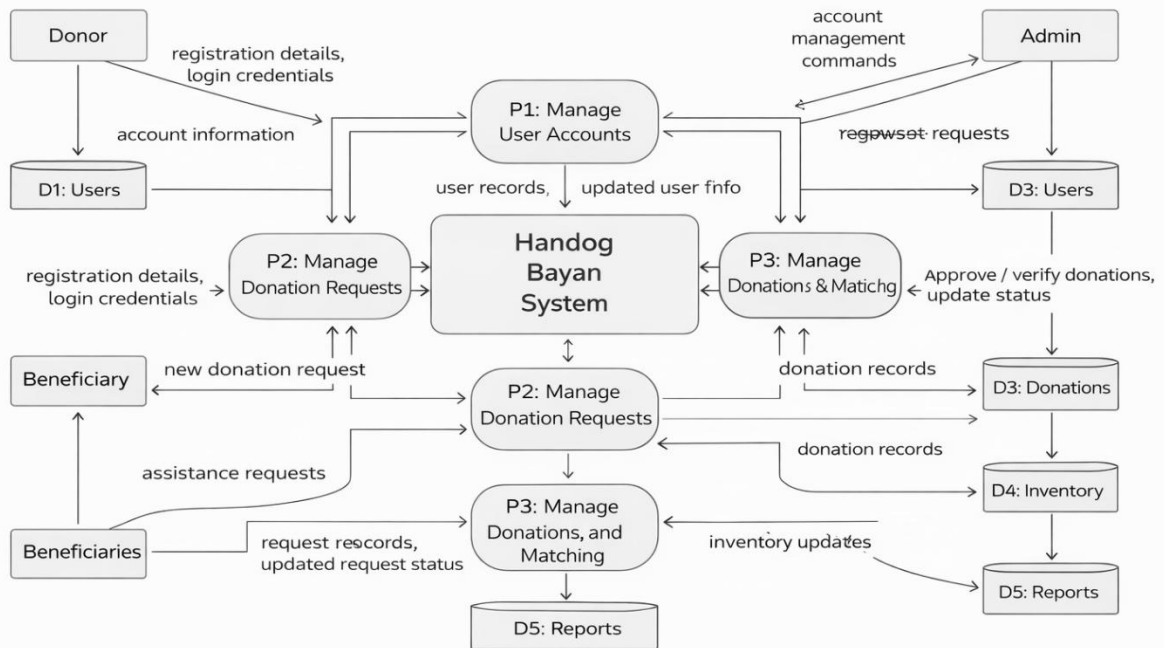
Use case



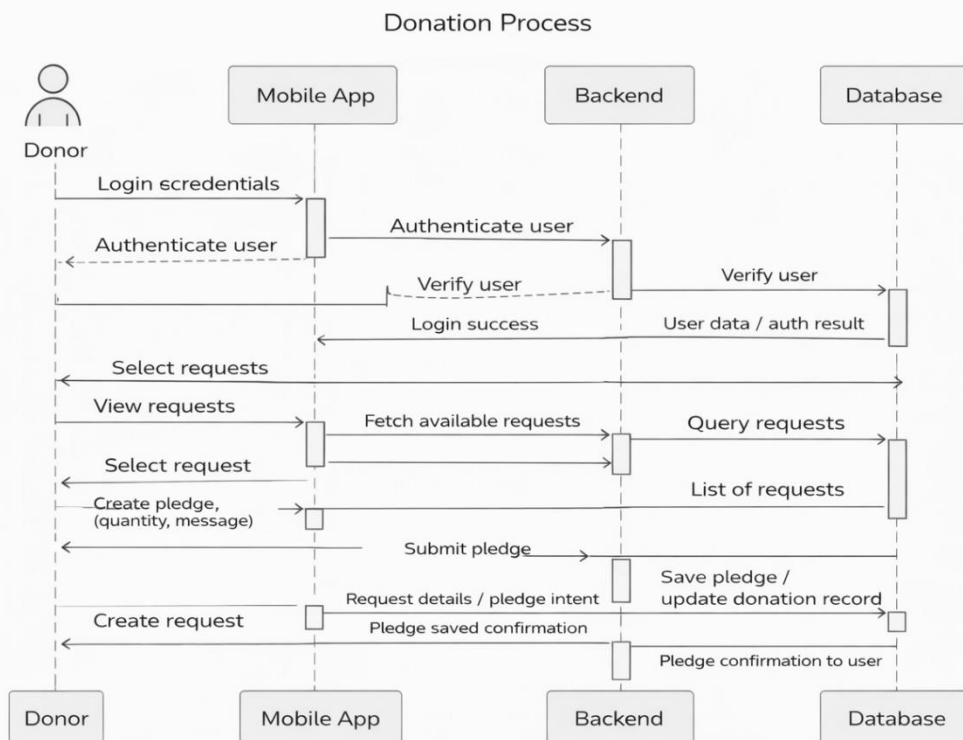
Context Diagram



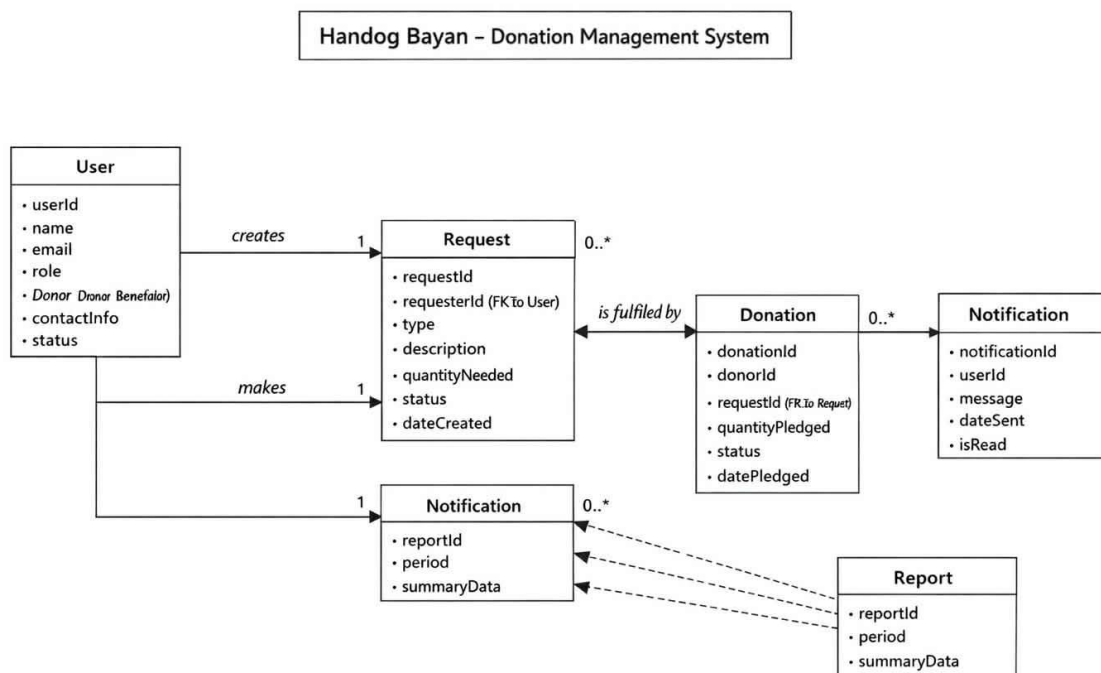
Data Flow Diagram (DFD – Level 1)



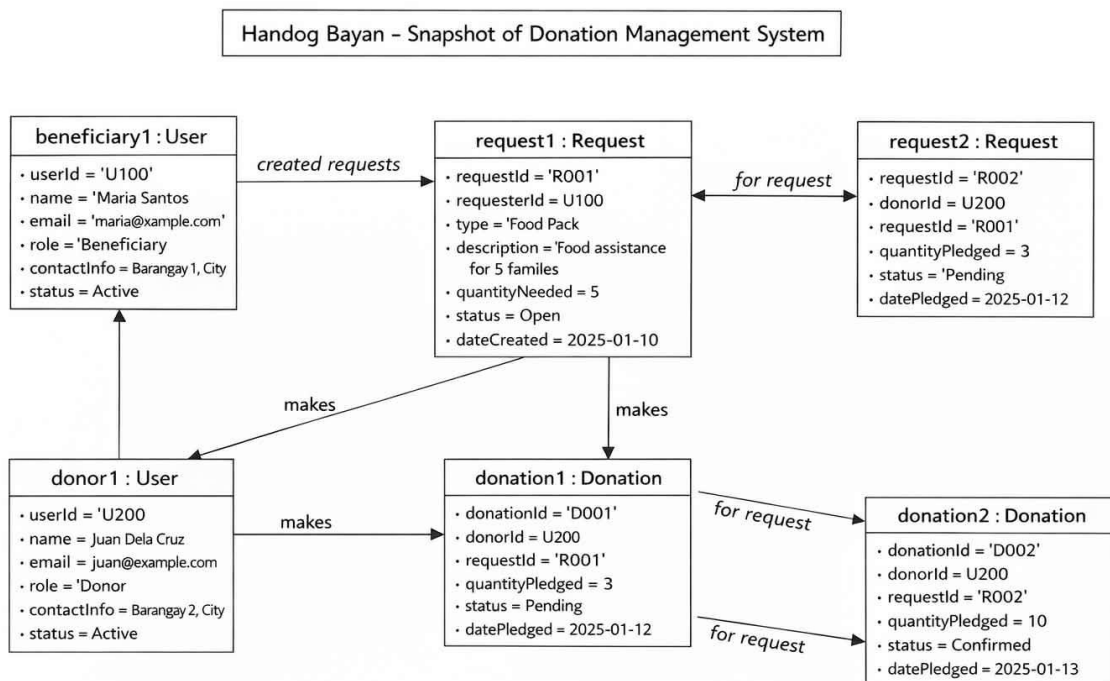
Sequence Diagram



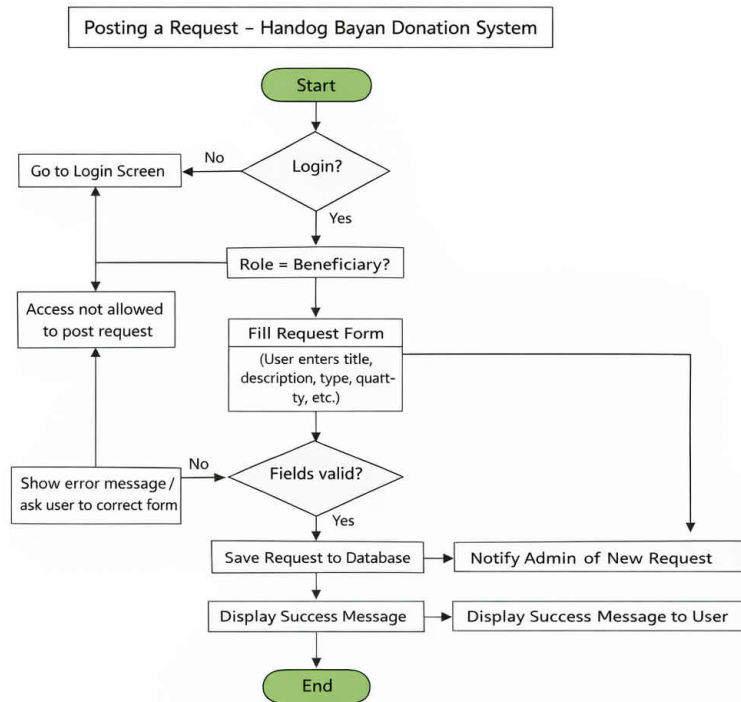
Class Diagram



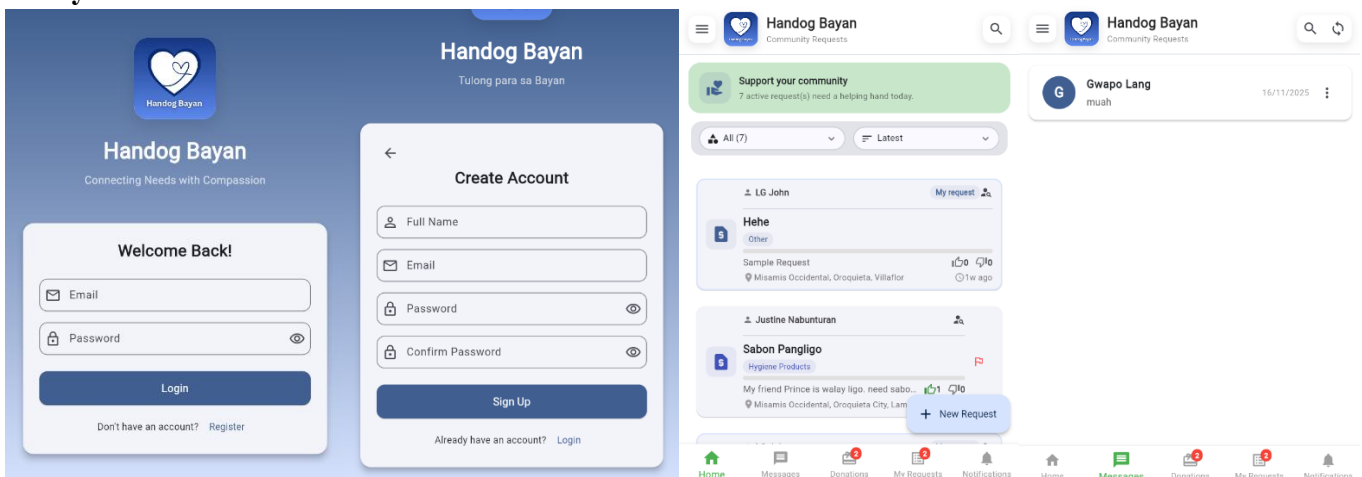
Object Diagram

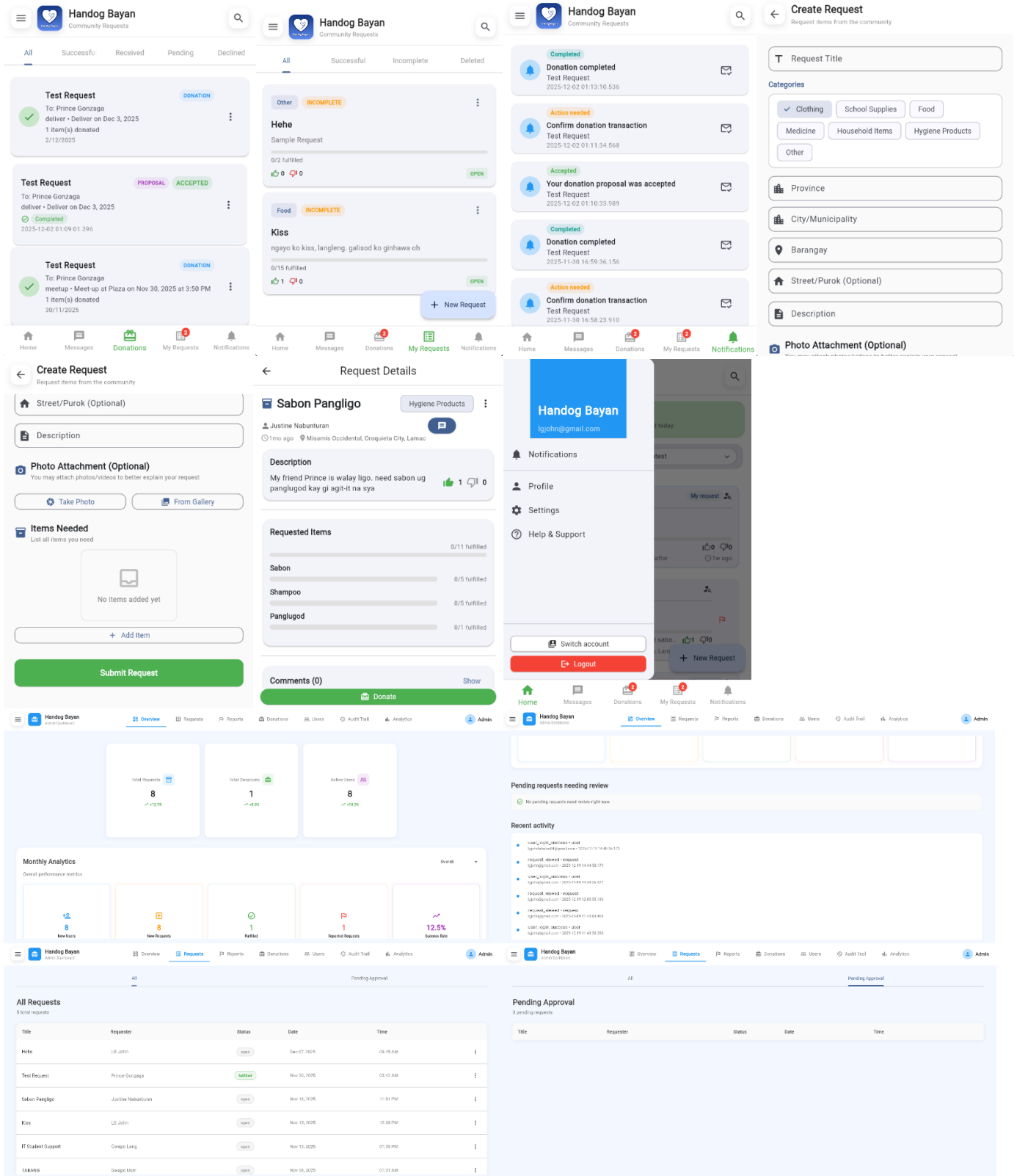


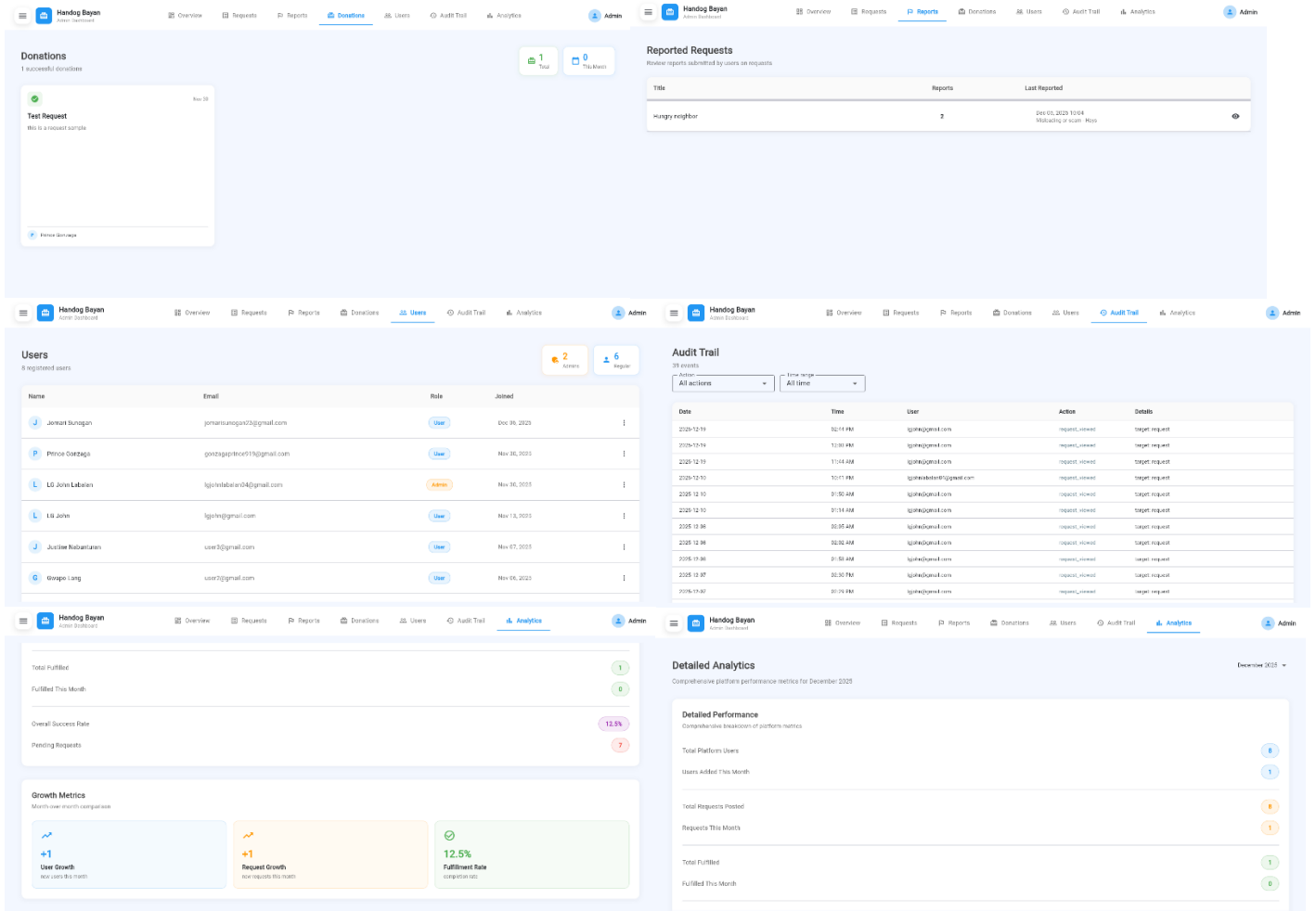
Flowchart



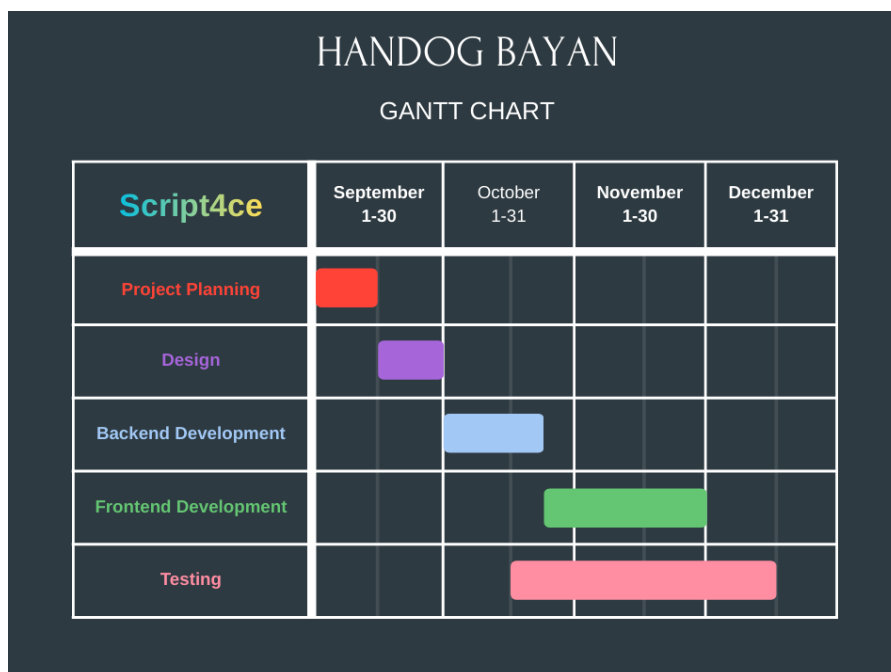
4.3 System Screenshots:







5.Calendar of Activities



6. Conclusion

The Handog Bayan system was conceptualized and developed to address persistent issues in traditional donation and relief operations, including fragmented communication, limited transparency, and inefficient tracking of donations. Through systematic requirement elicitation, structured design using UML and process diagrams, and iterative development, the project produced a working prototype that centralizes requests and donations and supports better coordination among donors, beneficiaries, and administrators.

Initial feedback from users suggests that the platform simplifies the process of finding legitimate needs, pledging support, and tracking the progress of donations. Moreover, the system's data storage and reporting capabilities lay a foundation for evidence-based planning of future relief and outreach activities.

Future work may focus on integrating secure digital payment systems, enhancing analytics (e.g., heat maps of needs, prediction of demand), improving offline capabilities for disaster scenarios with poor connectivity, and collaborating with government agencies and NGOs for broader adoption and sustainability.

Appendix A-References

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Appendix B - Data Gathering Evidence

- **Interview Guides and Transcripts**
 - List of questions asked to community organizers, donors, and potential beneficiaries.
 - Summaries or transcribed responses (with consent and anonymization where necessary).
- **Survey Instruments and Results**
 - Questionnaire forms
 - Summary tables and charts showing responses to key questions.
- **Observation Notes**
 - Field notes from observing donation drives or outreach events.
 - Photos of physical processes (if allowed) and brief descriptions.
- **Consent Forms and Ethics Documents**
 - Participant consent forms for interviews and surveys.

Appendix C - Project Planning

Provide **supporting documents and artifacts** related to project management, such as:

- **Detailed Gantt Chart** (by week) showing start and end dates for each task.
- **Work Breakdown Structure (WBS)** outlining major deliverables and sub-tasks.
- **Risk Management Plan** listing potential risks (e.g., delays in data gathering, technical issues) and mitigation strategies.
- **Meeting Minutes / Progress Reports** summarizing key decisions and milestones during development.

These documents demonstrate that the project was carried out systematically and within a planned timeframe.

Bibliography of The Researcher

I, **LG John Labalan**, am a third-year college student taking up the **Bachelor of Science in Information Technology** at the **University of Science and Technology of Southern Philippines**. With a strong interest in community service and the use of technology for social good, I conceptualized, designed, and developed the **Handog Bayan App** as an independent project.

As a **sole developer**, I performed all roles in the project, including the following:

- **Systems Analyst** – conducted background research, gathered user requirements, and defined the scope and objectives of the study.
- **Systems Designer** – designed the system architecture, including diagrams such as use case, context, data flow diagram (DFD), sequence, class, object, and flowchart, as well as the database structure.
- **Programmer** – implemented the Handog Bayan application, integrating the user interface with the backend and database.
- **Tester** – performed functional and user testing, identified and fixed defects, and refined system features based on feedback.
- **Documenter** – prepared the case study, technical documentation, and user-oriented system descriptions.

I aim to continue developing technology-based solutions that address real needs in Filipino communities, particularly in the areas of disaster response, donation management, and social welfare.